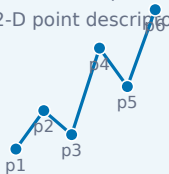


Observed anchors

M real GPS points
12-D point descriptors



Relational tokenization

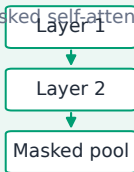
M point tokens
 $M(M-1)/2$ pair tokens



$$r_{ij} = \text{MLP}[x_i, x_j, x_j - x_i, \Delta t_{ij}]$$

Relation Transformer

2 layers / 4 heads
Masked self-attention



Trajectory representation

mean / std / max
statistics + quality
expert probability p_R



No coordinate interpolation | Long-range relations from real observations